# Bioenergetics

## Store light energy

### 1 Photosynthesis

**Carbon dioxide + water → glucose + oxygen.** Light transfers energy to chloroplasts; chlorophyll absorbs it.

**Endothermic:** energy is taken in. Plants respire all the time; photosynthesis needs light.

### 2 Use the glucose

Respiration; starch and fat/oil storage; cellulose cell walls.

Glucose plus nitrate-derived nitrogen → amino acids → proteins.

## Control & measure

### 3 Limiting factors

Relative light intensity: 4 → 1

Light, $CO_2$, temperature and chlorophyll can limit rate. A plateau means extra light alone will not help.

**Intensity ∝ 1/distance²;** doubling distance quarters intensity.

Greenhouses: balance extra crop income against heat/light/$CO_2$ costs.

### 4 Pondweed practical

Change lamp distance; measure oxygen volume/time. Control temperature, $CO_2$ supply, plant and background light.

Acclimatise; repeat; calculate mean. Bubbles vary in size, so volume is usually better.

Rate = volume/time; plot against light intensity or 1/distance$^2$.

## Release & use energy

### 5 Respiration pathways

**Aerobic:** glucose + $O_2$ → $CO_2$ + water; more energy transferred.

**Anaerobic:** muscles → lactic acid; yeast/plants → ethanol + $CO_2$. Less energy per glucose.

Respiration is exothermic and continuous; fermentation makes bread rise and produces alcohol.

### 6 Exercise & metabolism

Exercise raises heart rate, breathing rate/volume. Oxygen shortage → more anaerobic respiration and fatigue.

Oxygen debt: extra oxygen needed afterwards. Blood takes lactic acid to liver → glucose.

Metabolism = all cell/body reactions; builds carbohydrates, lipids and proteins; excess nitrogen → urea.